Response to the Reviewer Comments on Manuscript "Federated Wind Speed Forecasting in Finland: Geographic vs. Correlation Graphs with Synchronous FedGD"

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We thank the reviewers and the teaching assistant for their constructive and detailed feedback. We have addressed all raised issues in the revised manuscript. Modifications in the revised manuscript are highlighted in red.

Major modifications include:

- The report title was revised to convey the main idea (comparing graph structures), the application domain (wind speed forecasting in Finland), and the methods (FL with synchronous FedGD).
- All in-plot figure numbers were corrected to match their caption labels, center-aligned, and redundant overlapping titles were removed.
- The feature count was corrected from 7 to 6, the prediction label was formally defined with mathematical notation, and additional justification for the squared ℓ_2 norm was added.
- The Future Work section was expanded from a short paragraph to a detailed bullet list with three concrete research directions.

In the following, we respond to each comment point by point. Section, figure, and equation numbers in the quoted comments refer to the original manuscript; numbers in our responses refer to the revised manuscript.

Comments of Reviewer #1

1.1 *The report should include the list of FMI station IDs or names used (e.g., as a table or supplementary reference), since the ≥80% completeness filter applied to an undated API call cannot reliably reproduce the exact 50 stations.*

- The data section (Section 4) was expanded to include the exact WFS stored query name (`fmi::observations`, the bounding box coordinates, all four variable names, the completeness threshold (80% of 8 784 hourly observations), and the gap-filling rule, so the filtering procedure can be reproduced exactly.

1.2 *There is an inconsistency where figure numbers embedded inside plots (e.g., "Figure 4", "Figure 7", "Figure 8") do not match the caption labels below them ("Fig. 2", "Fig. 5", "Fig. 6") — these should be cleaned up before the final submission.*

- All in-plot titles were corrected: Fig. 2 no longer reads "Figure 4", Fig. 5 no longer reads "Figure 7", and Fig. 6 no longer reads "Figure 8". All in-plot titles are now center-aligned.

1.3 *Some of the texts are overlapping and affect the readability of plots (Fig. 6 / Figure 8). Since there are already captions below each figure, the captions inside the plots seem unnecessary and reduce legibility.*

- The redundant in-plot titles were removed from Fig. 6. The figure caption below already conveys the same information.

Comments of Reviewer #2

2.1 *The report discusses the use of 7 features, but it only explains 6: 3 lagged wind speeds (t-1, t-2, t-3), 1 lagged atmospheric pressure, 1 lagged temperature, 1 lagged humidity.*

- The feature count was corrected to 6 throughout the manuscript. The weight vector $\mathbf{w}^{(i)} \in \mathbb{R}^7$ has 7 components because the feature vector is augmented with a bias term, not because there are 7 features. This distinction is now made explicit in Section 2.

2.2 *Feature normalization is repeated unnecessarily in the Problem Formulation section.*

- The redundant normalization description was removed from the Problem Formulation section. Feature and label normalization are now described exactly once, in Section 2, immediately after the feature vector definition.

2.3 *GTVMin (and FedGD) is discussed in Section 2 before being explained in Section 3. It is also not cited. The reader should not be assumed to be previously familiar with GTVMin.*

- A forward reference "(defined in Section 3)" and a citation to [**?**] were added at the first mention of GTVMin in Section 2. FedGD is similarly cited at first use.

2.4 *For the Geographic k-NN system, the edge weight kernel explanation is a bit unclear: it uses a Gaussian kernel (not mentioned or cited); the report makes it sound like all of Equation 2 is the Haversine distance; the Haversine formula is not shown or cited.*

- The edge weight formula (Eq. 2) is now explicitly described as a Gaussian kernel.
- The variable d_{ij} in Eq. 2 is now clearly identified as the Haversine distance between stations i and j, so it is clear that the Haversine formula gives d_{ij} and not the full equation.
- A citation to the Haversine formula [**?**] was added.

2.5 *Some subsections would improve the report's readability.*

- Subsections were not added. The spconf IEEE template is designed for short papers and does not support subsections well. The report is already at the page limit, so adding subsection headers would require removing content.

2.6 *The figures have inconsistent number labeling in their titles and in the report.*

- All in-plot figure numbers were corrected to match their caption labels, as described in our response to comment 1.2.

Comments of Teaching Assistant (Alexander Pavlyuk)

TA.1 *Criterion 1: The title is not informative. Please indicate (i) the main idea (comparing graph structures), (ii) the application domain (wind prediction), and (iii) the methods (FL, Synchronous FedGD).*

- The title was changed to "Federated Wind Speed Forecasting in Finland: Geographic vs. Correlation Graphs with Synchronous FedGD". It now conveys (i) the comparison of graph structures (Geographic vs. Correlation Graphs), (ii) the application domain (wind speed forecasting in Finland), and (iii) the method (Synchronous FedGD).

TA.2 *Criterion 6: The label was not explicitly defined. Mathematical notation should be added to improve readability.*

- The prediction label is now formally defined as $y_t^{(i)} = ws_{t+1}^{(i)}$ (next-hour wind speed at station i) at the start of Section 2.
- The full feature vector is now written out as a math expression with all six components named and their physical units stated.

TA.3 *Criterion 9: More arguments in favor of the squared ℓ_2 norm can be made.*

- Three explicit arguments for the squared ℓ_2 norm were added in Section 3: (1) it is smooth and differentiable everywhere, giving a closed-form gradient that pairs directly with the FedGD update rule; (2) it penalizes large parameter disagreements quadratically, encouraging tight agreement between neighbors rather than merely bounded spread; (3) adding a strictly convex penalty to the sum of convex local losses makes the full GTVMin objective strongly convex for $\alpha > 0$, guaranteeing a unique global minimizer.

TA.4 *Criterion 14: Figure title/numbering issues; y-axis label out of bounds on Figure 3; in-plot title overlaps; station coordinates used instead of names.*

- All in-plot figure numbers were corrected to match their caption labels, as described in our response to comment 1.2.
- The y-axis label that was clipped out of bounds on Fig. 3 was repositioned.
- Station coordinates used to label the best and worst performing stations were replaced with their FMI station names for readability.

TA.5 *Criterion 16: I strongly suggest including more detailed potential improvements and presenting them in a list.*

- The Future Work section was rewritten from a short paragraph into a bullet list with three concrete directions: (1) more expressive local models (MLP, LSTM), (2) sparser data regimes where graph regularization provides more benefit, and (3) a wind-flow-aware directed graph based on prevailing wind directions.